Nonlinear control

Andrea Bisoffi

Department of Electronics, Information and Bioengineering
Politecnico di Milano

POLITECNICO
MILANO 1863

Academic year 2025/26

General information

Students:

- M.Sc. in Automation and Control Engineering
- M.Sc. in Mechanical Engineering
- M.Sc. in Mathematical Engineering

Teacher:

- Andrea Bisoffi
- Office: Campus Leonardo,
  Dept. Electronics, Information and Bioengineering (DEIB),
  building 20, floor 2
- Email: andrea.bisoffipolimi.it
- Office hours: to be arranged via email, possibly after lecture

General information

Students:

- M.Sc. in Automation and Control Engineering
- M.Sc. in Mechanical Engineering
- M.Sc. in Mathematical Engineering

Teacher:

- Andrea Bisoffi
- Office: Campus Leonardo,
  Dept. Electronics, Information and Bioengineering (DEIB),
  building 20, floor 2
- Email: andrea.bisoffipolimi.it
- Office hours: to be arranged via email, possibly after lecture

General information

- **Schedule**: Tuesday 08:30-10:15 room 26.0.3
  Friday 08:30-10:15 room E.P.7
- **Teaching mode**:
  - all classes take place in the lecture room **without** streaming in Webex.
  - All classes are recorded and the recording is posted after the lecture.
  - (Black)board except for the initial review.
- **Structure** of the course: 5 credits divided into
  - approx. 44 hours of lectures/exercises
  - approx. 6 hours of computer sessions

General information

- **Schedule**:

Tuesday	08:30-10:15	room 26.0.3
Friday	08:30-10:15	room E.P.7

- **Teaching mode**:
  - all classes take place in the lecture room **without** streaming in Webex.
  - All classes are recorded and the recording is posted after the lecture.
  - (Black)board except for the initial review.
- **Structure** of the course: 5 credits divided into
  - approx. 44 hours of lectures/exercises
  - approx. 6 hours of computer sessions

General information

- **Schedule**: Tuesday 08:30-10:15 room 26.0.3
  Friday 08:30-10:15 room E.P.7
- **Teaching mode**:
  - all classes take place in the lecture room **without** streaming in Webex.
  - All classes are recorded and the recording is posted after the lecture.
  - (Black)board except for the initial review.
- **Structure** of the course: 5 credits divided into
  - approx. 44 hours of lectures/exercises
  - approx. 6 hours of computer sessions

General information

- **Examination**: written exam with exercises and theory questions.
- **Teaching material**: slides and notes will be progressively uploaded to WeBeep.
- **Suggested textbooks**:
  Main textbook:
  - H. K. Khalil, "Nonlinear Control", Pearson, 2015

  Other textbooks:
  - C. M. Kellett and P. Braun, "Introduction to Nonlinear Control: Stability, Control Design, Estimation", Princeton University Press, 2023
  - H. K. Khalil, "Nonlinear Systems 3rd Edition", Prentice Hall, 2002
  - M. Vidyasagar, "Nonlinear systems analysis", SIAM, 2002
- **Prerequisites**: differential equations, multivariable calculus, linear algebra, main ideas from a B.Sc. course in control engineering.

General information

- **Examination**: written exam with exercises and theory questions.
- **Teaching material**: slides and notes will be progressively uploaded to WeBeep.
- **Suggested textbooks**:
  Main textbook:
  - H. K. Khalil, "Nonlinear Control", Pearson, 2015

  Other textbooks:
  - C. M. Kellett and P. Braun, "Introduction to Nonlinear Control: Stability, Control Design, Estimation", Princeton University Press, 2023
  - H. K. Khalil, "Nonlinear Systems 3rd Edition", Prentice Hall, 2002
  - M. Vidyasagar, "Nonlinear systems analysis", SIAM, 2002
- **Prerequisites**: differential equations, multivariable calculus, linear algebra, main ideas from a B.Sc. course in control engineering.

General information

- **Examination**: written exam with exercises and theory questions.
- **Teaching material**: slides and notes will be progressively uploaded to WeBeep.
- **Suggested textbooks**:
  Main textbook:
  - H. K. Khalil, "Nonlinear Control", Pearson, 2015

  Other textbooks:
  - C. M. Kellett and P. Braun, "Introduction to Nonlinear Control: Stability, Control Design, Estimation", Princeton University Press, 2023
  - H. K. Khalil, "Nonlinear Systems 3rd Edition", Prentice Hall, 2002
  - M. Vidyasagar, "Nonlinear systems analysis", SIAM, 2002
- **Prerequisites**: differential equations, multivariable calculus, linear algebra, main ideas from a B.Sc. course in control engineering.

General information

- **Examination**: written exam with exercises and theory questions.
- **Teaching material**: slides and notes will be progressively uploaded to WeBeep.
- **Suggested textbooks**:
  Main textbook:
  - H. K. Khalil, "Nonlinear Control", Pearson, 2015

  Other textbooks:
  - C. M. Kellett and P. Braun, "Introduction to Nonlinear Control: Stability, Control Design, Estimation", Princeton University Press, 2023
  - H. K. Khalil, "Nonlinear Systems 3rd Edition", Prentice Hall, 2002
  - M. Vidyasagar, "Nonlinear systems analysis", SIAM, 2002
- **Prerequisites**: differential equations, multivariable calculus, linear algebra, main ideas from a B.Sc. course in control engineering.

Questions?